

CODE:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, head, seek_time = 0;

    printf("Enter the number of disk requests: ");
    scanf("%d", &n);

    int request_queue[n];

    printf("Enter the disk request queue:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &request_queue[i]);
    }

    printf("Enter the initial position of the disk head: ");
    scanf("%d", &head);

    // Sort the request queue
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (request_queue[i] > request_queue[j])
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    {
        int temp = request_queue[i];
        request_queue[i] = request_queue[j];
        request_queue[j] = temp;
    }
}

// SCAN Scheduling
printf("\nSCAN (Elevator) Disk Scheduling:\n");

int start = 0;
int end = n - 1;
int current_direction = 1; // 1 = right, -1 = left
while (start <= end)
{
    if (current_direction == 1)
    {
        for (int i = start; i <= end; i++)
        {
            if (request_queue[i] >= head)
            {
                seek_time += abs(head - request_queue[i]);
                head = request_queue[i];
                start = i + 1;
                break;
            }
        }
    }
}

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        }
    }
    current_direction = -1;
}
else
{
    for (int i = end; i >= start; i--)
    {
        if (request_queue[i] <= head)
        {
            seek_time += abs(head - request_queue[i]);
            head = request_queue[i];
            end = i - 1;
            break;
        }
    }
    current_direction = 1;
}
}

printf("Total Seek Time: %d\n", seek_time);
printf("Average Seek Time: %.2f\n", (float)seek_time / n);
return 0;
}

```

OUTPUT:

Output

Enter the number of disk requests: 3

Enter the disk request queue:

123

234

342

Enter the initial position of the disk head: 234

SCAN (Elevator) Disk Scheduling:

Total Seek Time: 108

Average Seek Time: 36.00

=== Code Execution Successful ===